

Don't buy a fake. A stakeholder-ready view of the marketplace risk landscape.

Phase 1 turns the raw assignment into an executive narrative: where the strongest signals sit, where the biggest evaluation traps hide, and why a TabPFN v2 baseline is the fastest credible modeling move for this dataset.

11,358 TRANSACTIONS

10.7% FAKE RATE

522 EXACT DUPLICATES

The dataset is not huge, but it is operationally messy: repeated products, shortcut-heavy review text, and process metadata that can overwhelm more stable product-quality signals if we evaluate carelessly.

DATASET AT A GLANCE

The problem is manageable in size, but the data behaves more like a production system than a toy benchmark.

11,358

Marketplace transactions available for modeling

10.7%

Observed positive rate for the `Fake` target

19.1%

Rows with missing `top\_review`

9.7%

Rows with missing `Brand`

WHAT MATTERS IMMEDIATELY

- **Mixed modality:** numeric, categorical, date, and text all contribute signal.
- **Heavy repetition:** products recur so often that naive random splits would be misleading.
- **Missingness is informative:** process fields are not missing at random.
- **Shortcut risk is real:** review-star language and delivery metadata can dominate offline scores.

Fake Rate by Purchase Year

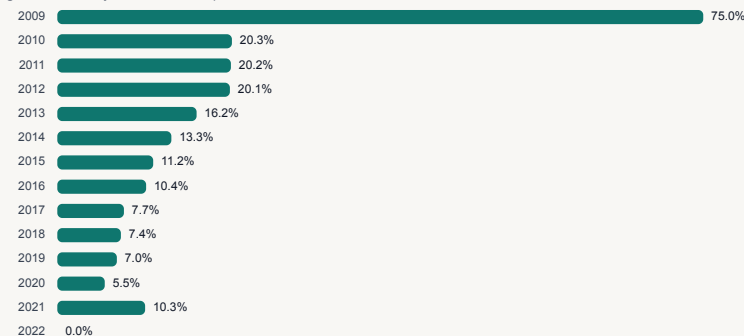
Older cohorts appear more fake-prone than more recent purchases.



Recent purchase cohorts are not uniformly safer; the portfolio still carries persistent fake exposure across years.

Fake Rate by Product Vintage

Older product vintages are materially more counterfeit-prone than recent cohorts.



Older product vintages are materially more counterfeit-prone than newer platform vintages.

The strongest predictors are useful, but some of them are dangerously close to workflow artifacts.

WHAT WE LEARNED

- **Review stars are explosive:** `1/5 stars` pushes fake rate to 89.1%.
- **Missing delivery data is risky:** overall fake rate jumps to 63.5% when delivery category is absent.
- **Pricing is not just “cheap = fake”:** aggressive overpricing relative to confirmed references is also suspicious.
- **Products repeat constantly:** the same names appear over and over, which can inflate apparent generalization.

BUSINESS IMPLICATION

A high-scoring model can still be brittle if it mainly learns workflow shortcuts. The job is not just to predict fake listings; it is to predict them for the right reasons and in a way that survives new products, new merchants, and new review patterns.

ENTITY LEAKAGE RISK

METADATA SHORTCUT RISK

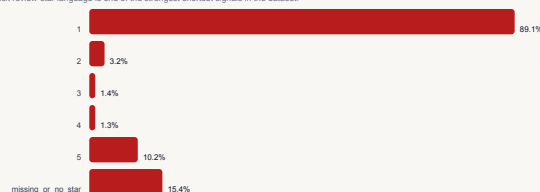
POLICY REVIEW NEEDED

Delivery category fake rate

Delivery metadata is not just predictive; it is potentially encoding process behavior.

Fake Rate by Review Star Bucket

Explicit review-star language is one of the strongest shortcut signals in the dataset.



Review-star text behaves like a near-shortcut and must be ablated later.

Price ratio fake rate

Listings above the confirmed-price reference look riskier than those near parity.

The single most important pattern in this dataset is not a standalone column. It is the interaction between **fulfillment mode** and **missing delivery metadata**.

BUSINESS + missing

80.0% fake rate across 155 rows. This is far beyond the base rate and likely reflects a workflow state, not just a product signal.

CENTRAL + missing

73.1% fake rate across 160 rows. The same pattern repeats in another operational channel, strengthening the shortcut hypothesis.

MARKETPLACE + missing

32.1% fake rate across 131 rows. Still risky, but notably less extreme than BUSINESS or CENTRAL.

Fulfillment delivery interaction

Ranked interaction slices highlight how quickly the risk spikes once delivery metadata disappears.

Fulfillment delivery heatmap

The heatmap makes the structure obvious: the missing-delivery state behaves very differently across flows.

This dataset can flatter a model very easily. The evaluation design matters as much as the algorithm choice.

WHY RANDOM ROWS ARE DANGEROUS

- **11,177 rows** belong to repeated product names.
- The same product families appear across both classes.
- Entity repetition means a random split can quietly teach the model product identity instead of counterfeit behavior.

RECOMMENDED EVALUATION POLICY

Area	Recommendation
Primary split	Use a temporal holdout on `PurchDate` instead of naive random rows.
Robustness check	Run a grouped split by `Brand + ProductName` to quantify entity leakage.
Cleaning	Remove exact duplicates and quarantine conflicting duplicate-like rows.
Primary metric	Use PR-AUC and operating-threshold precision/recall, not accuracy alone.

CORE TAKEAWAY

The model that wins this case study is not the one with the flashiest architecture. It is the one that keeps evaluation honest and avoids overclaiming on shortcut-heavy offline signals.

TabPFN v2 is the fastest credible baseline for this phase.

WHY TABPFN V2 FITS THIS CASE

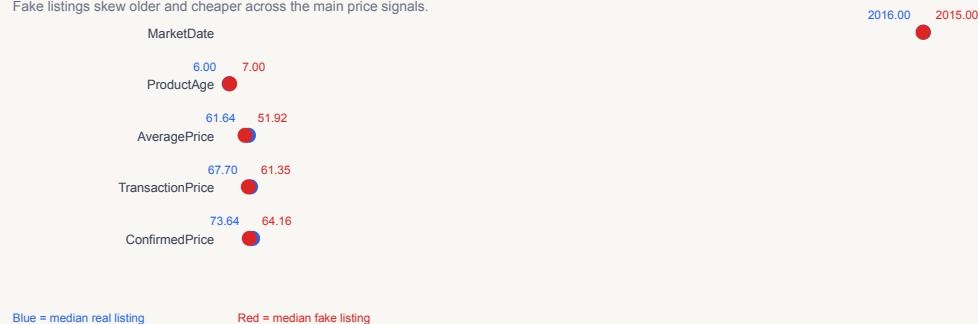
- **No long training cycle:** it gives us a strong foundation-model baseline immediately.
- **Great for a take-home:** small code surface, clear setup, and credible technical story.
- **Works with scalar features:** ideal once we convert text into lightweight review and length signals.
- **Good bridge strategy:** we can keep the PyTorch hybrid as the next upgrade only if TabPFN leaves headroom.

MVP FEATURE BLOCKS

Block	Examples
Numeric	prices, age, market year, date breakdown
Derived	price gaps, price ratios, missingness flags
Categorical	brand, city, zip, fulfillment, delivery
Text-derived scalar	review star value, lengths, keyword indicators

Median Numeric Features: Real vs Fake Listings

Fake listings skew older and cheaper across the main price signals.



The median comparison chart shows why a scalar-feature baseline is already informative.

IMPORTANT CAVEAT

Per TabPFN's official guidance, GPU is preferred for larger datasets, while CPU is best treated as a smoke-test path unless we explicitly force a full large-dataset run.

GPU PREFERRED

CPU SMOKE TEST

V2 WEIGHTS

The repository is already positioned as a stakeholder-ready package, not just a notebook dump.

EDA PIPELINE

`src.eda` generates tables and SVGs for missingness, temporal drift, pricing, review shortcuts, and interaction slices.

REPORTING LAYER

We ship both an executive summary and a technical report, plus this print-ready presentation deck.

MODELING MVP

`src.tabpfn\_pipeline` provides the minimum viable inference workflow around TabPFN v2.

RUNBOOK

Step	Command
Generate analysis artifacts	<code>python -m src.eda</code>
Build the markdown report	<code>python -m src.report</code>
Run TabPFN smoke test	<code>python -m src.tabpfn_pipeline --device cpu --max-train-rows 1000</code>
Run full TabPFN on stronger hardware	<code>python -m src.tabpfn_pipeline --device auto</code>

Use TabPFN v2 as the first serious model, but keep the governance lens on. This dataset contains enough shortcut signal that the evaluation policy is part of the model design.

1. Run the MVP

Score the cleaned temporal split with TabPFN v2 and inspect validation/test outputs.

2. Stress-test shortcuts

Re-run without review stars and without delivery metadata to measure reliance on brittle signals.

3. Upgrade only if needed

If TabPFN leaves room, move to the custom PyTorch hybrid already scaffolded in the repo.